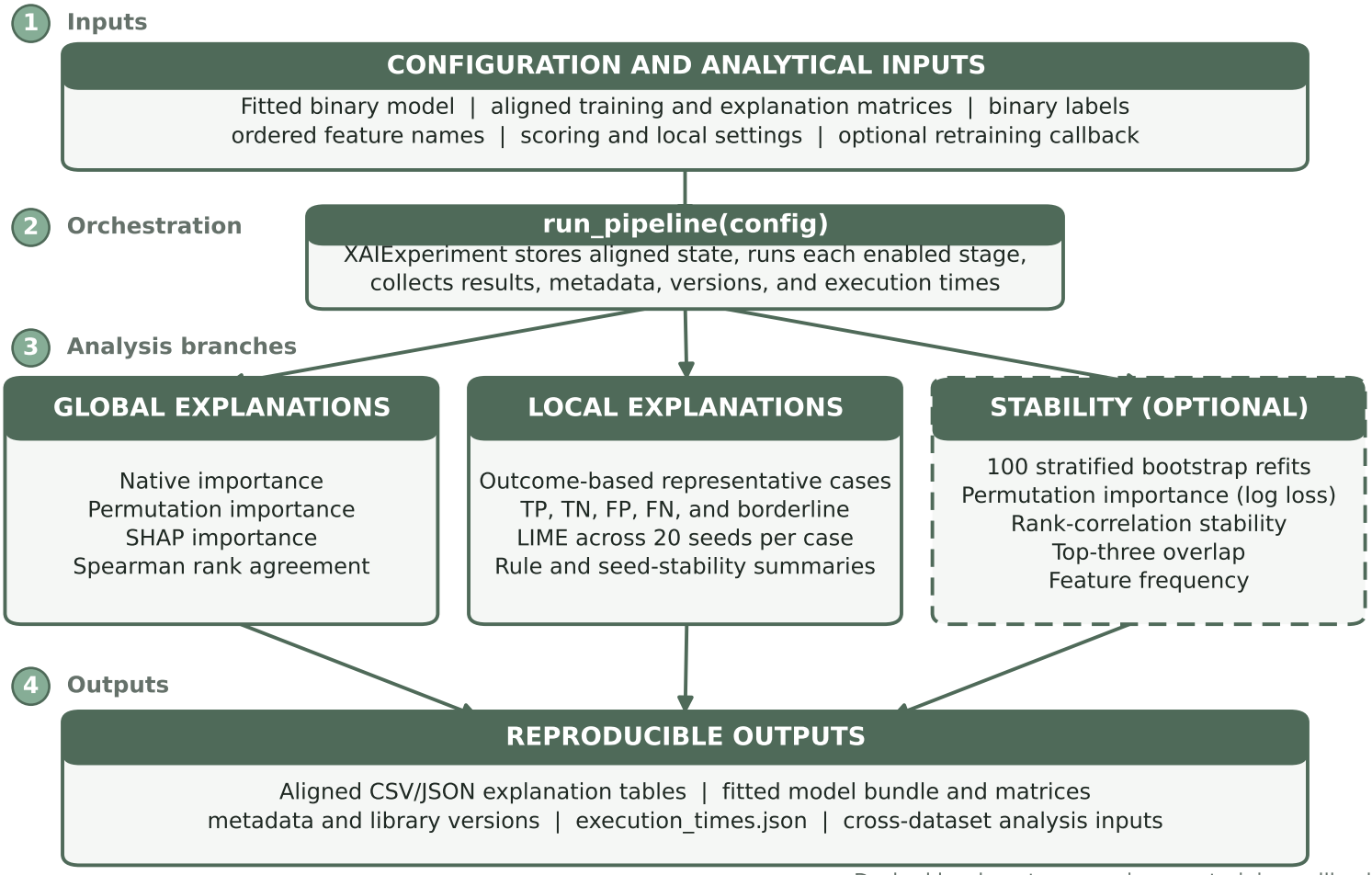
1
Inputs
CONFIGURATION AND ANALYTICAL INPUTS
Fitted binary model | aligned training and explanation matrices | binary labels
ordered feature names | scoring and local settings | optional retraining callback
2
Orchestration
run_pipeline(config)
XAIExperiment stores aligned state, runs each enabled stage,
collects results, metadata, versions, and execution times
3
Analysis branches
GLOBAL EXPLANATIONS
Native importance
Permutation importance
SHAP importance
Spearman rank agreement
LOCAL EXPLANATIONS
Outcome-based representative cases
TP, TN, FP, FN, and borderline
LIME across 20 seeds per case
Rule and seed-stability summaries
STABILITY (OPTIONAL)
100 stratified bootstrap refits
Permutation importance (log loss)
Rank-correlation stability
Top-three overlap
Feature frequency
4
Outputs
REPRODUCIBLE OUTPUTS
Aligned CSV/JSON explanation tables | fitted model bundle and matrices
metadata and library versions | execution_times.json | cross-dataset analysis inputs
Dashed border: stage requires a retraining callback